Activity 3.1: Audio Signal Extraction

Assignee: Humesh Sharma **Work Item:** Story **Allocated Duration:** 1h 30m **Status:** In Progress

1. Overview & Scope

In this activity, audio input was captured from a microphone or extracted from recorded video files associated with classroom sessions. The raw audio signal was processed to convert it into a digital waveform suitable for analysis. Sampling rate normalization and channel standardization were applied to ensure consistent processing across different audio sources.

The extracted audio signal was segmented into fixed-duration windows to enable time-based analysis aligned with video. These audio segments form the base input for noise and speech activity detection in subsequent stages. Processed audio data was stored in structured formats, including timestamps and amplitude information, to facilitate synchronized multimodal analysis.

2. Technical Specifications & Ingestion Parameters

Parameter	Configured Value	Design Justification
Sampling Rate	16,000 Hz (16 kHz)	Standard frequency for acoustic speech models and voice activity detection (VAD).
Channel Format	Single Channel (Mono)	Eliminates phase discrepancies and standardizes multi-mic classroom setups.
Quantization Depth	32-bit Float ([-1.0, 1.0])	Normalized from raw 16-bit PCM integer buffers to avoid numerical clipping during FFT.
Windowing Alignment	Fixed Time Windows	Enables precise timestamp correlation with facial engagement and attendance video frames.

3. Module Implementation (`backend/core/audio_analysis.py`)

```
backend/core/audio_analysis.py                                Python 3.10+
```

```
import librosa
 numpy  np
 os

 :
     (, sample_rate=):
        .sample_rate = sample_rate

     (, audio_data):

        :

             isinstance(audio_data, bytes):

                y = np.frombuffer(audio_data, dtype=np.int16).astype(np.float32) /
            :
                y = audio_data

             len(y) == :



             {: , : len(y), : .sample_rate}
         Exception  e:
             {: str(e)}importasimportclassAudioAnalyzerdef__init__self16000selfdefanalyze_
        Analyzes raw audio bytes/array:
        1. Noise level (decibels)
        2. Speech vs silence ratio
        3. Sound activity patterns

        Returns engagement metrics
        """try# Convert bytes to float32 numpy array if neededif# Assuming 16-bit PCM32768
```

4. Downstream Multimodal Pipeline Integration

> **Synchronous Alignment Note:** Audio chunks extracted in this unit pass through normalization and are correlated via ISO timestamps with concurrent frame outputs from the visual attendance tracker, driving integrated attention-index scoring across classroom sessions.